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# Comparison and uncertainty of multivariate modeling techniques to characterize used nuclear fuel

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ABSTRACT

The ability to characterize used nuclear fuel (UNF) is important for nuclear nonproliferation safeguards, criticality safety, and fuel storage. Multiple efforts have been made to estimate the burnup (BU), initial enrichment (IE), and cooling time (CT) based on multivariate models of isotopic concentrations and radiation signatures of the fuel. This work provides a comparison of multivariate modeling techniques and extends previous work by quantifying the uncertainty of the best model to predict each characteristic. Model inputs used are simulated gamma and neutron emissions from UNF of varying BU, IE, and CT. Modeling techniques explored include Ordinary Least Squares Regression (OLS), Principal Component Regression (PCR), and Partial Least Squares Regression (PLS). Multiple PCR and PLS models were built based on different variable selection methods, such as cross validation and Akaike Information Criteria. The OLS model predictions have a root mean square percent error (RMSPE) of less than 10%, but the models are very unstable. The PCR models exhibit a trade-off between accurate and stable predictions. The best performing PCR and PLS models have similar predictions errors, but the PLS models are favored due to their stability. The best model for each characteristic is a single output PLS model based on cross validation. The uncertainty of each of these models, based on their prediction variance and biases, is 0.220 GWd/MTU, 0.051% U-235, and 0.694 years for the BU, IE, and CT models, respectively. By building a 95% prediction interval based on the corresponding uncertainty of each characteristic, 1.97% of the BU predictions, 23.03% of the IE predictions, and 100% of the CT predictions lack 95% confidence that they are within the prescribed accuracy requirement for the characteristic.

## 1. Introduction

Characterization of used nuclear fuel (UNF) can inform decisions about multiple aspects of UNF handling, including nuclear nonproliferation safeguards, criticality safety, and storage of the fuel. Important fuel characteristics include the discharge burnup (BU), initial enrichment (IE), and cooling time (CT). The BU and IE provide information about the fissile material content and criticality safety and the CT provides information about decay heat of the fuel. There are a variety of nondestructive assay (NDA) measurements based on gamma and neutron signatures that can be used to verify the BU and CT of UNF [1,2]. These measurements rely on the ability to detect the emissions from and estimate the concentration of a select few isotopes, such as Cs-137 and Eu-154, that have known relationships to the BU and CT of the fuel [3]. It may be possible to verify the IE of UNF using self-interrogation neutron resonance densitometry [4].

These NDA measurements and correlations rely on a very small amount of information contained in the emissions from UNF. There is also a known non-uniqueness to the solution space when isotopic concentrations are used to characterize UNF [5] that decreases the accuracy of these correlations, especially at longer cooling times. To explore how to improve the accuracy and increase the amount of information gleaned from the emissions of UNF, others have used multivariate modeling to characterize UNF. Efforts have been made to model UNF characteristics, including the reactor type, based on isotopic concentrations [6,7]. These efforts used an unrealistic model input space, over 20 isotope concentrations [6], that is not readily observable during NDA measurements of UNF. Previous efforts have also shown very poor characterization of older UNF (14.3 year CT) [7], limiting the usefulness of such models to only recently discharged UNF. Simulated gamma spectra have also been used to predict these characteristics [8], which addresses the need to extract isotopic concentrations from NDA measurements. Using gamma spectra to model the characteristics shows promising results in the accuracy of the predictions based on gamma spectra. However, this work considered a very limited population of UNF characteristics and therefore does not fully capture how accurately this model input can predict the desired characteristics. Previous work explored incorporating neutron signatures into the model inputs with

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gamma signatures to improve the performance of the fuel characterization [9]. Including the early die away time (a neutron related signature) with the fission product activity and total Cherenkov light intensity (gamma related signatures) improved the accuracy of the predictions.

Understanding the accuracy and applicability of multivariate models for safeguards applications includes uncertainty analysis of the predictions to provide a prescribed confidence in the result. Work on the sensitivity of previous modeling efforts [7,9] shows how the models are impacted by different types of uncertainty. Procedures to investigate the model sensitivity include adding 1%–10% Gaussian noise into the data or modifying model parameters and comparing the results with the initial results. Adding random noise to the data and modifying model parameters explore how these models are affected by random noise, but do not yield the uncertainty in the predicted values or show how the predictions with uncertainty compare to required accuracy of each characteristic.

The previous work in modeling UNF characteristics suggests that multivariate modeling can be an accurate way to characterize UNF, but they each primarily use the same modeling techniques with little comparison of techniques and little effort to understand their uncertainties. This work explores and compares the performance of three different parametric modeling techniques and the uncertainty of the best model for each characteristic. Modeling techniques used include Ordinary Least Squares Regression (OLS), Principal Component Regression (PCR), and Partial Least Squares Regression (PLS), as these are commonly used techniques in fuel parameter modeling [6–8]. A variety of PCR and PLS models are developed based on multiple ways to select the number of latent variables to use. Performance metrics of interest include accuracy and stability of the models. The models use inputs of simulated gamma and neutron emissions from UNF to predict the discharge BU, IE, and CT of the fuel. The comparison of the models will highlight the advantages and disadvantages of each technique, and suggest the best way to model the UNF characteristics. The uncertainty analysis of the best model will provide insight on how well the predicted values are known and how the uncertainty compares to suggested accuracy limits.

## 2. Methodology

The gamma and neutron emissions of the fuel, the model inputs, were simulated using Origen 6.2 [10] based on a Westinghouse 17x17 PWR fuel assembly. All of the simulations consisted of 3 irradiation cycles of length 540 days at 100% power, with the power level varying based on the specified assembly-averaged discharge BU of the fuel. The first two cycles are followed by 30 days of down time to represent a normal refueling outage, and the third cycle is followed by the specified length of CT. This single simulated operation history provides a realistic simulation of operations at a plant in the U.S. A thorough investigation of the effect of operating history is necessary for deployment of these tools but is beyond the scope of this analysis. The resulting emissions were normalized to 1 MTU. The gamma emissions consist of 1024 logarithmically spaced energy bins from 100 keV to 4 MeV. The neutron emissions are collected into a single energy bin for energies of 1 eV to 1 MeV to yield a gross neutron emission count. The radiation emissions used are the emitted counts, as opposed to the measured counts. Therefore, the gamma spectrum and neutron counts are of higher intensity than what would be measured, and do not have any detector noise or detector effects. Investigating how detector effects impact the model inputs and model performance lies outside of the scope of this work and is left for future work.

The output space used for this work was selected based on common operation of nuclear power plants in the U.S. The BU space ranges from 20–70 GWd/MTU, in increments of 5 GWd/MTU. The IE of the fuel is calculated based on the BU value according to Eq. (1) [11], which is an empirical relationship with non-negligible error between the two fuel parameters assuming the reactor operators are running the reactor to most efficiently use the fuel.

$$IE = 0.31 * BU^{0.65} \tag{1}$$

Using this relationship provides a realistic characteristic space, as it is very rare to have combinations of high BU with lowly enriched fuel and vice versa when the reactor is under normal operating conditions. However, using this relationship between the two output variables also limits the ability of the models to independently characterize the IE of the fuel. To account for uncertainties in the coefficients of Eq. (1), provide robustness for the assumptions of the equation, and account for normal process variation, a range of $\pm$ 5 and 10% of the calculated IE values are also used. This leads to each BU value being paired with five realistic values of IE. The CT space consist of 5, 6, 7, 8, 9, 10, 12.5, 15, 20, 25, and 30 years. The bin sizes for the CT varies based on the exponential nature of radioactive decay, and include CTs of fuel that would be removed from a spent fuel pool.

Based on these parameters, the input data is a 605 $\times$ 1025 matrix, with each bin of the gamma spectra and the neutron counts as the input variables ($605 = 11 * 5 * 11, 1025 = 1024$ gamma energy bins + 1 gross neutron count). The output matrix is 605 $\times$ 3, with each of the desired characteristics as the output variables. The data set is divided randomly into three sets: a training set with 302 observations, a testing set with 151 observations, and a validation set with 152 observations. The training and testing sets are used to build and optimize models using each of the regression techniques. The results of the predictions of the testing data set are used to compare the models. The predictions of the validation data set are used to calculate the uncertainty of the best model.

Multivariate modeling techniques used in this work are OLS, PCR, and PLS. These are techniques that have been used in a variety of previous modeling efforts [6–8]. Multiple methods to determine the number of variables are used with PCR and PLS, such as cross validation (CV) [12] and Akaike Information Criteria (AIC) [13]. CV is the comparison of how a loss function changes as a function of a desired model parameter. The goal of CV is to select the model parameter value that minimizes a specified loss function. However, in comparing the loss function to a model parameter, user opinion of a visual inspection of the CV plot may suggest a different parameter value if the difference in the loss function is small. The inclusion of user opinion introduces considerations of the simplicity, or parsimony, of the model in addition to the fit, based on intuition that empirical models should be as simple as possible. When using CV, this work will compare the RMSPE of the predictions of the testing data set since the RMSPE of the training data when using CV is biased to be lower than expected on an unseen data set.

AIC is an information theoretic measure of the complexity of a model and describes the difference between estimated and true probability functions [13]. This method automatically balances the fit and parsimony of a model by taking into account the amount of variance lost by not including each latent variable (LV) and the degrees of freedom. AIC is a specialized form of CV, in that the model parameter that results in the lowest AIC value should be selected.

The models are evaluated based on their root mean square percent error (RMSPE) and the condition number. The RMSPE is used to estimate how accurately the model can predict the output of each observation. It is calculated based on Eq. (2), in which $y$ is the actual data and $\hat{y}$ is the predicted data.

$$\text{RMSPE} = \frac{\sqrt{\text{mean}((y-\hat{y})^2)}}{\text{mean}(y)} \times 100 \tag{2}$$

A percent error is used for equal comparison across each of the outputs, since they are on different scales. The RMSPE value for each model should be as low as possible. For safeguards purposes, there is a desire to measure the plutonium content in UNF to within 5% of the actual value, which corresponds to estimating the BU of the fuel

to within 5% [14]. Since the IE is related to the BU in this work, this goal is also applied to the IE RMSPE. Accuracy of the CT within ± 1 year allows for the discharge cycle of the fuel to be determined [5], therefore the CT should consistently be predicted within ± 1 year of the actual value. These accuracy requirements will be used in addition to the RMSPE value to evaluate the accuracy of each of the models.

The condition number of a model is used to evaluate the stability of each model. This metric describes how small perturbations in the data ($y$) are amplified in predictions ($f$), as shown in Eq. (3).

$$\frac{\|\delta f\|}{\|f\|} \leq \text{Condition Number} \times \frac{\|\delta y\|}{\|y\|} \tag{3}$$

The calculation of a condition number can be performed in a variety of methods. It is generally calculated as the ratio of the largest singular value of the matrix being inverted in linear regression to the smallest singular value of that matrix. Since the singular values of a matrix are related to the eigenvalues of a matrix, the condition number can also be calculated as the ratio of the maximum eigenvalue to the minimum eigenvalue of that matrix, as shown in Eq. (4). If a subset of the variables are used in the model, such as in PCR, then the eigenvalues used for this calculation are adjusted to only consider the variables used in the model.

$$\text{Cond} = \frac{\lambda_{max}}{\lambda_{min}} \tag{4}$$

The minimum value of a condition number is 1, which corresponds to the inversion of a scalar value. It represents that small changes in the input data set will cause at most an equal change in the output predictions. Large condition numbers represents that small changes in the input data will lead to large differences in the fitted parameters, which can lead to unstable or unrepeatable models. Therefore, large condition numbers are not favored. For this work, an absolute maximum threshold of 100 will be placed on the condition number for a model to be considered stable. This is based on common practices of empirical modeling.

To investigate the application of this methodology to the characterization of UNF, the uncertainty of the best model to predict each characteristic is quantified based on the predictions of the validation data set. Uncertainty quantification is a critical need in applications where decisions are made based on model predictions, such as in international safeguards. Evaluating the uncertainties of the best models provides the smallest range over which there is confidence in the predicted value. Uncertainty of multivariate models consists of three parts. The first is the noise variance, $\sigma^2_{noise}$, which comes from the noise in the measurement of the input and output variables. The second part is the prediction variance, $\sigma^2_{\hat{y}}$, which is the expected variance between the actual and predicted outputs. This is calculated through Monte Carlo sampling of the observations to build prototypic models. 500 prototypic models of 100 observations each are used to calculate the prediction variance, with the 95th percentile of the calculated variances used as a conservative estimate of the model prediction variance. The third part is the bias, which is the average difference between the actual and predicted values. These values are summed in accordance with Eq. (5) to yield the total uncertainty of the model in units of the characteristic being predicted.

$$\text{Uncertainty} = \sqrt{\sigma^2_{noise} + \sigma^2_{\hat{y}} + \text{bias}^2} \tag{5}$$

The total uncertainty is then used to build a 95% prediction interval (PI), which corresponds to ±1.96 times the total uncertainty. The PI provides the interval over which there is 95% confidence of the predicted value. When applied to the accuracy requirements previously stated, the PI will describe if there is 95% confidence that the requirement is met.

## 3. Ordinary least squares regression

The first multivariate technique used in this work is OLS. OLS uses the best linear combination of the input variables, $X$, to predict the output, $Y$, with error, $e$. To solve for the regression coefficients, $\beta$ in Eq. (6), a pseudoinverse of the input matrix, $X^T * X$, is used, as shown in Eq. (7) [15]. The use of a pseudoinverse allows for the coefficients to be solved for when the input matrix is not square or full-rank, which are conditions of matrix inversion.

$$Y = X * \beta + \epsilon \tag{6}$$

$$\beta = (X^T X)^{-1} X^T Y \tag{7}$$

While OLS is implied to perform best when there is a linear relationship between the inputs and outputs, OLS can still produce accurate predictions as long as the model is linear in parameters. This has previously been shown when quadratic and interaction terms are included in the model [16,17]. For this work, the input data set is not expected to have a perfectly linear relationship to the output data. Multiple isotopes, like Cs-134 and Eu-154, do not have a linear relationship to the BU of the fuel [3], which corresponds to nonlinearities between the gamma spectrum energy bins and the BU. The relationship between BU and IE in this work means that nonlinear relationships between the input variables and the BU hold true between the input variables and the IE. The CT is known to have an exponential decay relationship with radiation emissions. It is also desired in OLS models to have input variables that are uncorrelated to each other, to prevent the matrix from containing linear dependencies. In this work, it is known that the energy bins of the gamma spectrum are highly correlated due to multiple emissions and decay chains of single isotopes. The neutron emissions are correlated to some of the gamma energy bins due to gamma emissions during spontaneous fission, the primary source of neutrons in UNF.

The full input data set cannot be used to perform this OLS since it is not an invertible matrix. The input variables were down selected based on their correlation to other input variables. Input variables that have a correlation coefficient less than 0.99 to each of the previous variables was kept for this regression type. For example, for variable 3, the correlation coefficient to variable 1 and the coefficient to variable 2 were each examined to determine if they are below the threshold. The 0.99 threshold was selected because it removes perfectly correlated input variables (signified by a correlation coefficient of 1), while keeping enough data to still perform the regression and make predictions.

This process left 12 energy bins of the gamma spectrum as variables. Two of the notable energy bins remaining include the 511 keV photon from positron annihilation and 662 keV photon from the decay of Cs-137, which is known to have a linear relationship to the BU of the fuel [1]. The remaining energy bins are all below 0.9 MeV, with three energy bins being between 0.662 MeV and 0.9 MeV. These three energy bins would be the best resolved from a physical measurement, indicating that this model uses additional energy bins that may not be well-resolved in a physical measurement. The collapsing of the data helps to limit the linear dependency and correlation between input data. The substantial reduction in the size of the input data strongly suggests that a dimensionality reduction technique would perform better since it will be able to utilize more of the original input space. However, OLS is still used in this work with the collapsed input data set because of the simplicity of the regression.

## 4. Principal component regression

The next modeling technique used in this work is PCR. PCR uses a linear transformation of the input data, $x_j$, into the Principal Component (PC) space by multiplying each input variable by a loading vector, $p_{ji}$, to result in the scores for that PC [18], as shown in Eq. (8). The

loading vectors are the eigenvectors of the covariance matrix of the input data.

$$T_i = \sum_{j=1}^{n} x_j * p_{ji} \tag{8}$$

Each PC maximizes the amount of variance of the original data set while maintaining orthogonality to all of the previous PCs, and all of the PCs are ordered by decreasing amount of variance and information. There are a variety of methods to determine how many PCs should be used based on the amount of information abut the original data explained by each PC. This work uses five different methods to determine the number of PCs to use: enough PCs to explain at least 95% of the variance, PCs with eigenvalues greater than or equal to 1, PCs with correlation coefficients of at least 0.575 (signifying some linear correlation) to each output, AIC, and CV. A further discussion of how each of these models was developed is contained in [19].

## 5. Partial least squares regression

The third modeling technique used in this work is PLS. PLS is a supervised linear transformation of the input and the output data into a latent variable (LV) space. It involves two different linear transformations: an outer transformation and an inner transformation. The outer transformation is used to transform the inputs and outputs into a latent variable space, as shown in Eqs. (9) and (10).

$$X = t_i * p_i + E_i \tag{9}$$

$$Y = u_i * q_i + F_i \tag{10}$$

In these equations, $q_i$ and $p_i$ are the loadings for output and input latent variables, $u_i$ and $t_i$ are the score vectors on the loadings, and $E_i$ and $F_i$ are the residual errors for each LV. Multiple algorithms have been developed to calculate these loadings and scores such that they maximize the correlation between the input and output space. This work primarily uses the SIMPLS algorithm [20] to perform the outer transformation. The loadings are ordered by decreasing amount of the covariance that they explain. The outer transformation collapses correlated information into the same LV, so that each of the loadings of the input and output are not correlated, but not necessarily orthogonal as in PCR.

The inner transformation is used to relate the input and output score vectors in the LV space. This is often done by performing a linear regression on the scores using regression coefficients $\beta$ as shown in Eq. (11) in which $\epsilon_i$ is the error, but can be done using any form of regression.

$$u_i = t_i * \beta_i + \epsilon_i \tag{11}$$

The inner transformation utilizes a single LV at a time, resulting in a single input, single output regression.

For this technique, two primary methods are applied to build four different models. The first model predicts all three outputs and uses LVs informed by CV. Since PLS is a supervised transformation it is expected that changing the output data of the model will result in changes in the performance of the model. The next set of models each predicts a single output and use LVs informed by AIC. The third set of models predicts a single output and is informed by CV. The fourth model uses a nonlinear inner transformation and predicts only the CT, based on a nonlinear relationship between the input and output scores of the first LV to predict CT. Neural networks are used to perform the inner transformation, based on [21] and use LVs informed by CV. This last model uses the simplified PLS algorithm [15] to perform the outer transformation due to the ease of altering the inner transformation. The SIMPLS and simplified PLS algorithm have been shown to produce the same results when using a single output variable [20], therefore the change in algorithm does not introduce additional errors in the predictions. A further discussion of how each of these models was developed is contained in [19].

**Table 1**
Summary of results of each model to predict BU.

| Model | No. variables | RMSPE | Condition number |
|---|---|---|---|
| OLS | 12 | 3.51 | $9.35 \times 10^{37}$ |
| PCR, Variance | 1 | 31.27 | 1 |
| PCR, Eigenvalues | 4 | 7.45 | 150.8 |
| PCR, Correlation | 1 | 26.59 | 1 |
| PCR, AIC | 14 | 0.61 | $1.72 \times 10^{8}$ |
| PCR, CV | 17 | 0.59 | $7.03 \times 10^{8}$ |
| PLS, Multilinear | 13 | 0.57 | N/A |
| PLS, AIC | 25 | 0.57 | N/A |
| PLS, CV | 13 | 0.57 | N/A |

**Table 2**
Summary of results of each model to predict IE.

| Model | No. variables | RMSPE | Condition number |
|---|---|---|---|
| OLS | 12 | 4.32 | $9.35 \times 10^{37}$ |
| PCR, Variance | 1 | 21.96 | 1 |
| PCR, Eigenvalues | 4 | 8.23 | 150.8 |
| PCR, Correlation | 1 | 19.16 | 1 |
| PCR, AIC | 14 | 1.78 | $1.72 \times 10^{8}$ |
| PCR, CV | 19 | 1.72 | $8.83 \times 10^{8}$ |
| PLS, Multilinear | 13 | 1.69 | N/A |
| PLS, AIC | 25 | 2.08 | N/A |
| PLS, CV | 13 | 1.64 | N/A |

**Table 3**
Summary of results of each model to predict CT.

| Model | No. variables | RMSPE | > ±1 yr | Condition number |
|---|---|---|---|---|
| OLS | 12 | 8.55 | 41.7% | $9.35 \times 10^{37}$ |
| PCR, Variance | 1 | 48.36 | 90.7% | 1 |
| PCR, Eigenvalues | 4 | 20.48 | 70.9% | 150.8 |
| PCR, Correlation | 1 | 48.36 | 90.7% | 1 |
| PCR, AIC | 14 | 4.72 | 6.62% | $1.72 \times 10^{8}$ |
| PCR, CV | 14 | 4.72 | 6.62% | $1.72 \times 10^{8}$ |
| PLS, Multilinear | 13 | 4.60 | 7.28% | N/A |
| PLS, AIC | 25 | 8.11 | 17.9% | N/A |
| PLS, CV | 13 | 4.53 | 7.28% | N/A |
| PLS, Nonlinear | 11 | 10.77 | 53.0% | N/A |

## 6. Prediction results and comparison

The results of each model, including the number of variables used, are summarized in Tables 1–3, separated by the output being predicted. For most of the models, the BU predictions have the lowest RMSPE, followed by the IE then the CT. This is not surprising since there is at least one known linear relationship between a gamma signature and the BU, the 662 keV peak from Cs-137 decay. The IE values are based on the BU values, and therefore maintain some of the relationships between the signatures and the BU but with introduced nonlinearities based on Eq. (1). This highlights the dependency of the BU and IE values in this work. The CT is known to have a nonlinear relationship to radiation emissions based on exponential decay laws, and is therefore expected to contain the worst predictions from linear models.

The OLS model is able to provide more accurate predictions than expected, given that the data used is still very correlated. This suggests that there may be a stronger linear relationship between the inputs and the outputs than initially estimated. Further investigation should be performed as to how the variable down selection affected these results, and how another down selection method or subset of the input data would perform in predicting the output variables. These promising prediction results come at the expense of a very high condition number. The high linear correlation in the input data means that OLS will likely never yield a stable condition number for this input data set. Therefore, a transformation of the data is needed to remove the correlations.

The PCR models show a trade-off of stability and low RMSPE values and are never able to achieve both. This is primarily due to most of the variance of the input data being in the same direction, leading to most of the information of the input data captured in the first PC and little

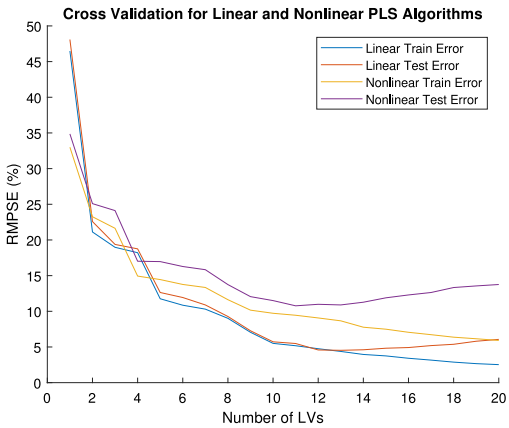


**Fig. 1.** Cross Validation of linear and nonlinear models to predict CT.

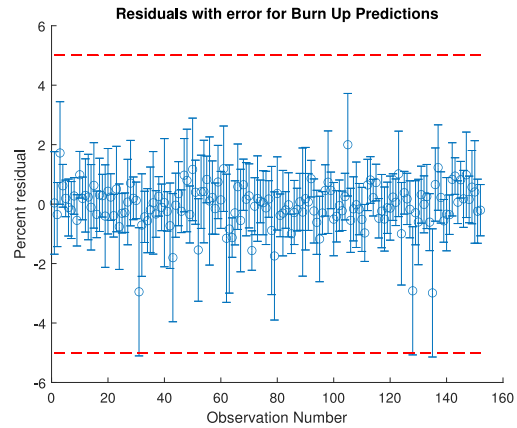


**Fig. 2.** Burnup residuals and PI for each validation observation compared to the 5% accuracy goal.

information in the others, This distribution of the variance causes the first PC to have a very high eigenvalue and the later PCs to have low eigenvalues. The information in the later PCs is needed for accurate predictions of the output data, causing the use of PCs with high and low eigenvalues and the high condition number of the models. These results also show that the PCR models provide the most accurate predictions when information in the input and output data are considered in the model selection, such as through AIC and CV.

The PLS models all have similar low RMSPE values and are all considered stable. The condition number of each of the PLS models is trivially 1, due to use of a single input, single output regression for the inner transformation. This means that the input score matrix is size $n \times 1$ and the matrix being inverted ($t_i^T * t_i$) is a 1x1 matrix, or a single value. Therefore the condition number does not change for each of the PLS models and has little physical meaning to the results of the model. For this reason, the condition number is presented as 'N/A' in Tables 1–3. The nonlinear PLS model to predict CT does not have a condition number since the neural networks use a scaled conjugate gradient descent to find the optimal parameters instead of matrix inversion.

The accuracy of the PLS models is similar to the PCR models built on CV and AIC. This shows that these two techniques are able to yield similar accuracy, with the main difference in their performance lying in the condition number of the models. Although the multilinear and CV PLS models to predict CT result in the lowest RMSPE values, the AIC and cross validation PCR models result in the lowest percent of observations predicted more than 1 year from the actual value.

With regards to the models to predict the CT, it is very surprising that the nonlinear model did not result in a lower RMSPE value than the linear model. The CV of each model was compared in Fig. 1 to determine why this occurs. The nonlinear model has a significantly lower RMSPE than the linear model when only 1 LV is used. This suggests that the nonlinear fit of the first LV is able to capture more information about the system. However, when the second LV is used as well, the linear model shows a much larger decrease in the RMSPE than the nonlinear model, and continues for the next few LV. This suggests that more information is obtained from the second LV in the linear model than the nonlinear model. These observations suggest that by placing more information about the system in the first LV of the nonlinear model, the subsequent LVs do not contain as much information as they do in the linear model and are adding noise into the regression. This leads to the higher overall RMSPE of the predictions.

Between all of the models, the PLS CV model is selected as the best. For each of the characteristics it results in the lowest RMSPE value of the stable models. These models are used for uncertainty analysis based on the predictions of the validation data set.

## 7. Uncertainty analysis

The total uncertainty and 95% PI of the PLS CV models were quantified to understand the confidence with which the predictions meet the specified accuracy requirements. Since all of the data in this work was simulated and there is no associated instrument noise, the noise variance is assumed to be 0 for each of the models. The 95th percentile of the prediction variances correspond to standard deviations of 0.219 GWd/MTU, 0.0510% U-235, and 0.692 years for the BU, IE, and CT models, respectively. The calculated biases are 0.0009 GWd/MTU, −0.0033% U-235, and 0.048 years for the BU, IE, and CT models, respectively. Combining these together using Eq. (5), the total uncertainties for these models are 0.220 GWd/MTU, 0.051% U-235, and 0.694 years for the BU, IE, and CT predictions, respectively.

The PI for each of the characteristics is ±0.4312 GWd/MTU, 0.100% U-235, and 1.360 years for the BU, EN, CT respectively. Comparing the differences between the actual and predicted values (residuals) and PI with the accuracy requirements can provide insight into the probability of false alarms. False alarms arise when the model predicts a value outside the required accuracy due to model inaccuracies instead of errors in the declared value. Figs. 2, 3, and 4 show the potential for a false alarm with respect to each characteristic. These figures show the residual of each observation, either the percent (BU and IE) or absolute (CT), with the PI as the error bar for each residual, and the accuracy criteria (the red dotted line). The residual values are compared with the accuracy criteria to determine how well the predicted values match the declared values, and how often there is 95% confidence that the predicted value is within the criteria.

The BU residuals, Fig. 2, show that all of the predicted values are within 5% of the actual value, a false alarm rate of 0%. Most of the predictions have 95% confidence to be within the acceptance criterion. but there are three observations that do not have 95% confidence that they are within 5% limit. This corresponds to a false alarm rate of 1.97%. The IE residuals, Fig. 3 shows one observation predicted outside the 5% limit, a false alarm rate of 0.66%. However, there are 35 observations that do not have 95% confidence that they are within the 5% limit, a false alarm rate of 23.03%.

Finally, the CT residuals, Fig. 4, shows that 19 observations are predicted outside the 1 year limit, a false alarm rate of 12.5%. None of the observations have 95% confidence that the predicted value is within 1 year of the actual value, a false alarm rate of 100%. This is due to the relatively large uncertainty of the CT predictions causing a PI greater than 1 year, so no future prediction will have 95% confidence of being within 1 year of the actual value. If the accuracy crtieria is relaxed to 2 years, which still limits the possible discharge cycle of the fuel and provides useful information to inspectors, only 51 of the observations do not have the 95% confidence. This is a false alarm rate of 33.55%.

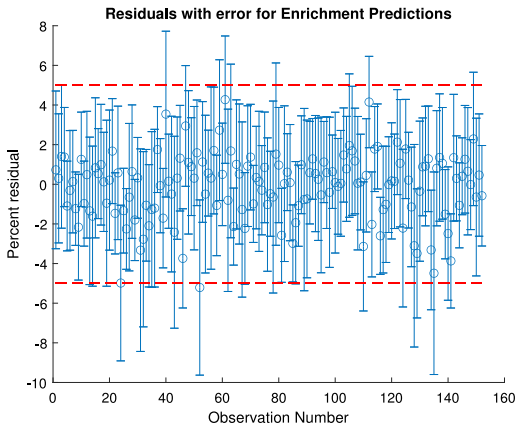


**Fig. 3.** Enrichment residuals and PI for each validation observation compared to the 5% accuracy goal.

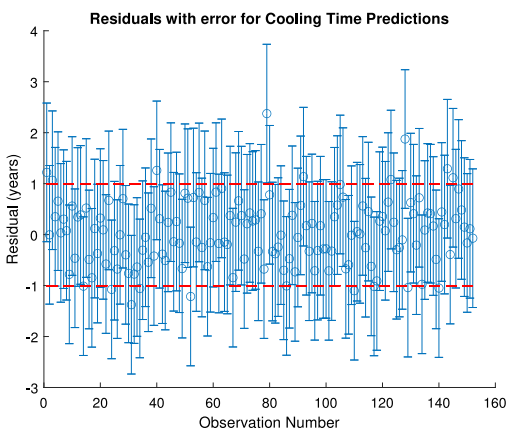


**Fig. 4.** Cooling time residuals and PI for each validation observation compared to the 1 year accuracy goal.

## 8. Conclusions

To accurately characterize UNF based on relatively easy to measure NDA signatures, multivariate modeling has been investigated. This work built and compared multiple models to characterize the BU, IE, and CT of UNF based on simulated gamma and neutron emissions of Westinghouse 17x17 PWR fuel. The PLS models are able to consistently produce the most accurate results, while PCR models are either accurate or stable. The uncertainty of the best model to predict each characteristic was calculated to better understand the confidence of the predicted values. The BU model was shown to predict within 5% of the actual value with 95% confidence in almost all predictions. The CT model is never able to predict within 1 year of the actual value with 95% confidence due to the large uncertainty of the predictions. About two-thirds of the predictions show 95% confidence of being within 2 years of the actual value.

The largest limitation of this work is the use of simulated radiation emissions as the model inputs, as opposed to the detected radiation. The emissions are not able to be measured in a physical system, contain more photopeaks than would be present in a measured gamma spectrum, and do not contain detector effects, such as a Compton continuum. The differences in the gamma spectrum contribute to the accuracy and relatively low uncertainty of the predictions. It is suggested that the inputs of the models be changed to measured radiation counts to provide a model based on measurable signatures.

Additionally, this work only considers one operation history. Changes in the operation history, including extended outages or forced outages, affects the concentration of short-lived isotopes (such as Cs-134) in the fuel and the corresponding signal in the radiation signatures. The predictions from these models will not be as accurate if the conditions of the training data samples are not upheld. A logical extension would be to compare predictions of observations of different operation history to explore at what point changes in the operation history causes significant changes in the predictions and build other models to capture the effects of the operation history on the radiation signatures.

An additional extension of this work would be to incorporate the noise variance into the uncertainty. If simulated signatures continue to be used as inputs, the noise variance would come from the uncertainties in SCALE. If the input data is changed to measured radiation counts, the noise variance would come from the detector and instrument noise. Adding the noise variance would increase the total uncertainty, PI, and false alarm rate of each model.

## CRediT authorship contribution statement

**Amanda M. Bachmann:** Methodology, Formal analysis, Writing - original draft. **Jamie B. Coble:** Conceptualization, Methodology, Supervision, Writing - review & editing. **Steven E. Skutnik:** Methodology, Writing - review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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